FARAMADE MICHAEL AYENI

Applied AI & Intelligent Systems | Computer Science
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RESEARCH PROFILE

First-Class Computer Science graduate (CGPA 4.84/5.00) with hands-on experience spanning low-resource speech recognition research and full-stack software engineering. Also contributed to the design, implementation, training, and evaluation of TARA, a tone-aware Yorùbá language automatic speech recognition system based on Conformer-CTC. Interested in building deployable AI-enabled systems that transform data, language, and domain knowledge into useful decision-support and intelligent applications.

RESEARCH INTERESTS

Applied Artificial Intelligence • Intelligent Systems • Machine Learning • Speech & Language Technologies • Low-Resource AI • AI-Enabled Decision Support

EDUCATION

Bachelor of Science (B.Sc.) in Computer Science | *Kwara State University, Malete, Nigeria* **2023 - 2026**
First Class | CGPA: 4.84/5.00

- Relevant coursework: Artificial Intelligence; Data Mining & Data Warehousing; Analysis of Algorithms; Data Structures & Algorithms; Linear Algebra; Statistics; Numerical Methods; Distributed Systems; Database Systems.

RESEARCH EXPERIENCE

Undergraduate Researcher - TARA: Tone-Aware Recognition and Annotation | *Department of Computer Science, Kwara State University* **2025 - 2026**
Final-Year Research Project | Supervisor: Dr. J. F. Ajao

- End-to-end development of a tone-aware Yorùbá language automatic speech recognition system using a Conformer-CTC architecture, with core focus on speech-data preparation, model implementation/training, and experimental evaluation.
- Prepared and structure a custom corpus of 1,691 validated Yorùbá recordings from five speakers; standardized audio to 16 kHz mono WAV and preserved tone marks, under-dotted characters, and other orthographic distinctions in a 94-character output vocabulary.
- Configured and trained a 6-block Conformer-CTC model using Python, PyTorch, NVIDIA NeMo, and PyTorch Lightning with AdamW optimization, cosine-annealing learning-rate scheduling, and CTC loss.
- Evaluated the held-out test set using Word Error Rate (45.0%), Character Error Rate (24.0%), and Diacritic Error Rate (32.5%), and analyzed error patterns associated with limited low-resource speech data.
- Integrated the trained model into a transcription interface that accepts Yorùbá audio and returns tone-aware text output.

SELECTED AI & TECHNICAL PROJECTS

AcademicGPS - AI-Assisted Academic Performance & Planning Platform **2026**

- Developing an intelligent academic support platform that analyzes student academic records to surface performance trends, CGPA progression, and personalized planning insights.

Market Intelligence System - AI-Enabled Decision Support **2026**

- Developing an AI-assisted market intelligence system designed to transform market and business information into structured insights that support analysis and decision-making.

AI Portfolio Generator & Hosting Platform **2026**

- Developing an AI-assisted platform that captures career information, identifies strengths, generates tailored professional portfolio content, and supports shareable hosted portfolio experiences.

TechSphere - Full-Stack Web Publishing Platform **2025 - 2026**

- Built a full-stack technology publishing platform with authentication, session management, author-level CRUD controls, rich-text editing, MongoDB persistence, and external technology-news API integration using Node.js, Express.js, EJS, MongoDB/Mongoose, Quill.js, and JavaScript.
  Live: techsphere-7fak.onrender.com

PROFESSIONAL EXPERIENCE

Software Development Intern (SIWES) | *Lagos State Ministry of Innovation, Science and Technology - Application Department, Oracle Unit* **1 Sep 2025 - 10 Feb 2026**

- Completed structured full-stack engineering training and practical application development using JavaScript, Node.js, Express.js, EJS, MongoDB/Mongoose, SQL/PostgreSQL, REST APIs, Git/GitHub, and Postman.
- Developed web applications including TechSphere, a weather forecasting application, a lost-and-found logging system, and an interactive decision-based game while applying authentication, database persistence, API consumption, and responsive UI principles.
- Strengthened debugging and systems-thinking skills by resolving authentication mismatches, database connectivity issues, template-rendering errors, and external API failures in iterative development workflows.

TECHNICAL SKILLS

Programming: Python; JavaScript (ES6+); TypeScript; SQL; HTML5; CSS3
AI / Machine Learning: PyTorch; NVIDIA NeMo; PyTorch Lightning; Conformer-CTC; supervised ML experimentation; speech-data preprocessing; ASR evaluation (WER, CER, DER)
Web / Systems: React; Node.js; Express.js; REST API integration; EJS; responsive web development
Databases: MongoDB; Mongoose; PostgreSQL
Tools: Git; GitHub; Postman; Visual Studio Code; Google Colab

LEADERSHIP & TECHNICAL COMMUNITIES

Core Team Member | *Google Developer Groups on Campus (GDGoC), Kwara State University* **2024 - 2026**

- Participated in developer-community activities, including the Build with AI event, and engaged with student-focused technical learning and collaboration initiatives.

Community Member | *RAIN-IN KWASU* | *beTechified Community*

- Participated in technology-focused learning communities and professional-development activities; also attended FoundersConnect organized by Hult Prize KWASU.

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

The Complete Full-Stack Web Development Bootcamp | *Udemy - Angela Yu*
Introduction to Career Skills in Software Development | *LinkedIn Learning* Mar 2025
Data Science & Analytics | *HP LIFE* Oct 2024
Introduction to Cybersecurity Awareness | *HP LIFE* Oct 2024
Phone Software: Hacks and Configuration | *Nigerian Youth Academy* Apr 2025